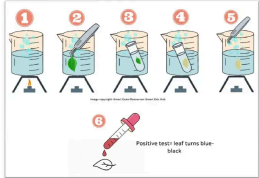


CAMBRIDGE LOWER SECONDARY
STAGE 9- BIOLOGY

Topic: Photosynthesis
• **Why** 06 Know that photosynthesis occurs in chloroplasts and is the process by which plants make carbohydrates, using the energy from light.
• **Why** 07 Know and use the summary word equation for photosynthesis (carbon dioxide + water → glucose + oxygen, in the presence of light and chlorophyll)
Photosynthesis:
• It is the process by which plants make carbohydrates, using the energy from light.
Site of photosynthesis:
• Photosynthesis occurs in chloroplasts.
Word equation for photosynthesis:
• **Word equation for photosynthesis is:**
carbon dioxide + water → glucose + oxygen, in the presence of light and chlorophyll

Experiments:
Experiment 1 (Test for carbohydrates):
Testing for carbohydrates involves a colour change if starch is present. Since chlorophyll is a green-coloured pigment, it might prevent the colour change from being noticed. Hence it becomes essential to remove chlorophyll from the leaves.
Process overview:
Killing the cells in the leaf → Removing Chlorophyll → Testing for starch
Diagram (A&B)



Method of killing the cells in the leaf:
• **Step 1:** Water is boiled in a beaker and then the flame is switched off.
• **Step 2:** Using a pair of forceps, the leaf is kept submerged in the beaker containing hot water for around 30 seconds. This results in killing the cells.
• When the cells are killed, they are unable to react any further. Secondly, it allows the iodine solution to enter the cells easily.
Method of removing chlorophyll from the cells:
• **Step 3:** The leaf with dead cells is placed in a test tube. The test tube is then covered with ethanol. The test tube is then placed in the beaker of hot water.
• Ethanol has a very low boiling point as compared to water.
• The hot water in the beaker is enough to boil the ethanol.
• **Step 4:** As a result, chlorophyll from the leaves dissolves in the ethanol in the test tube, at the same time, makes the leaf brittle.
• **Step 5:** The brittle leaf is then removed from the test tube using forceps and dipped in hot water to make it softer and easier to handle for the next steps.

Method of testing the leaf for the presence of starch:
• **Step 6:** The soft leaf is then placed on a white tile.
• It is then stained with iodine solution.
• A positive test for starch, changes the colour of starch grains to blue-black

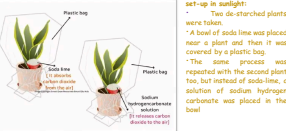
Experiment Summary:
• The leaf is dipped in hot water to kill the cells.
• The leaf is then submerged in boiling ethanol to extract the colour from the leaf.
• The ethanol will change to a green colour while the leaf should lose all of its colour to become white.
• When brown-coloured iodine solution is added to the leaf, and if it turns to a dark blue-black colour, then it indicates the presence of starch in the leaf.

Experiment 2 (To check if carbon dioxide is needed for photosynthesis):
Process overview:
De-starch the plant → Leaving the plant under special set-up in sunlight → Testing for starch

Note:
• If you want to check if carbon dioxide is necessary for photosynthesis, then you need to remove the starch that is already present in the leaves. This is called as de-starching a plant.

Method of de-starching a plant:
• Soak the plant in dark for 2/3 days

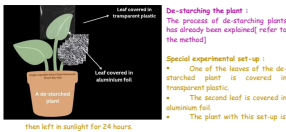
Process of checking if carbon dioxide is needed for photosynthesis:



Note:
• The purpose of using soda lime is to eliminate any CO₂ in the air, and the purpose of using sodium hydrogen carbonate was to add CO₂ to the air.
• Both were covered up by plastic bags to restrict the environment to a smaller space so that the soda lime and sodium hydrogen carbonate can be more of effective.

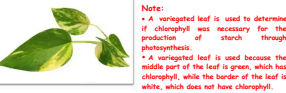
Testing for starch:
• After 24 hours, starch test was performed for the plant that had soda lime. This should test negative for starch.
• While the plant that had sodium hydrogen carbonate should test positive for starch, indicating that photosynthesis had taken place and starch had been produced.

Experiment 3 (To check if light is needed for photosynthesis):
Process overview:
De-starching the plant → Special experimental set-up → Testing for starch



Testing for starch:
• The leaf covered in transparent plastic gives a positive test for starch.
• The leaf covered in aluminum foil gives a negative test for starch.

Experiment 4 (To check if chlorophyll is needed for photosynthesis):
Process overview:
Using a variegated leaf → Testing for starch

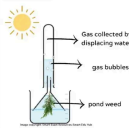


Using a variegated leaf:
• The plant with variegated leaves is left in sunlight for 24 hours.
Observations:
• The plant with variegated leaves that is left in sunlight for 24 hours, still has variegated leaves.
Testing for starch:
• The iodine test is performed on one of the variegated leaves.
• Only the part where there were chlorophyll cells will turn to a blue-black colour while the rest of the leaf will not change colour.

Experiment 9 :To check if oxygen is produced during photosynthesis:

Process overview:

Using a Canadian pondweed -> Testing the gas collected for oxygen



Testing the gas collected for oxygen:

- The gas produced can be tested using a glowing splint. The glowing splint relights, confirming the presence of oxygen gas.

The gas collected for oxygen

The experimental set-up:

- Two identical Canadian pondweed plants are used and placed in a setup as shown in the diagrams above.
- One is placed in light and the other is placed in dark for 24 hours.
- Observation:
- Bubbles are seen rising from the funnel in the test tube kept in sunlight, indicating the formation of some gas.
- The gas gets collected over water by displacing the water.

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CAMBRIDGE LOWER SECONDARY
STAGE 9- BIOLOGY

Topic: Genetics:

Species:

A species is a group of organisms that can breed to produce a fertile off-spring.

Example:

Horse and donkey are not the same species. They both do breed together to produce an offspring called a mule. But the mule is infertile. As per the definition, if the offspring as a result of breeding is infertile then the two animals that bred together belong to different species.

Variation within a species:

Variation can be a result of genes or the environment.

Variation:

- Variation is the difference between the individuals in a species or groups of organisms belonging to the same species.
- Variation within a species is related to genetic differences between them.

There are two types of variations:

- Continuous variation
- Discontinuous variation:

Continuous variation:

- Continuous variation is a variation in which a series of intermediate types connects the extremes.

Example 1: Difference in the height

Example 2: Difference in the length of fingers

Example 3: Difference in the skin colour

Discontinuous variation:

Discontinuous variations are caused by sudden unpredictable changes in the chromosome or gene mutation.

Example 1:

Blood groups in humans show discontinuous variation. The parents change in the blood group is only caused by his genes and not his environment.

Example 2 : Eye colour

Example 3: Labeled or labelless ears

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99p.03 Describe variation within a species and relate this to genetic differences between individuals.

Inherited characteristics:

- 99p.03 Know that chromosomes contain genes, made of DNA, and that genes contribute to the determination of an organism's characteristics.

Inheritance:

Chromosomes/Sex chromosomes/Fertilization:

Chromosomes:

- Chromosomes are thread-like structures located inside the nucleus of animal and plant cells.
- Each chromosome is made of protein and a single molecule of deoxyribonucleic acid (DNA).
- Different species have a different number of chromosomes. Example: Humans have 22 pair of chromosomes, horses have 32 pairs of chromosomes.

Sex chromosomes:

- A sex chromosome is a type of chromosome involved in sex determination.
- Two of the chromosomes (the X and the Y chromosomes) determine your sex as male or female when you are born. They are called sex chromosomes.
- Females have 2 X chromosomes.
- Males have 1 X and 1 Y chromosome.

Gene:

- Genes are segments of deoxyribonucleic acid (DNA) that contain the code for a specific protein that functions in one or more types of cells in the body.

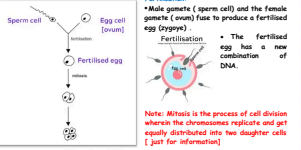
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Life processes

- 99p.01 Describe the fusion of gametes to produce a fertilised egg with a new combination of DNA.
- 99p.02 Describe the inheritance of sex in humans in terms of XX and XY chromosomes.
- 99p.04 Describe the scientific theory of natural selection and how it relates to genetic changes over time.

Gametes:

- Gametes are sex cells.
- In humans sperm is the male sex cell or male gamete and ovum is the female sex cell or female gamete.
- Gametes are produced by a special type of cell division (called as meiosis - you need not know the name, useful at this stage of learning; but you need to know what happens as a result of this process). In this type of cell division, new cells are produced that are genetically different. During this type of cell division, the chromosome number is halved.
- Sex cells (sperm and ovum) are also produced by meiosis.



Fertilisation:

- Male gamete (sperm cell) and the female gamete (ovum) fuse to produce a fertilised egg (zygote).
- The fertilised egg has a new combination of DNA.

Note: Meiosis is the process of cell division wherein the chromosomes replicate and get equally distributed into two daughter cells (just for information)

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Inheritance of sex:

- Humans have 23 pairs of chromosomes.
- 22 pairs are autosomes (body chromosomes) and the 23rd pair is the sex chromosome.
- The sex chromosomes are referred to as X and Y, and their combination determines a person's sex.
- Typically, human females have XX chromosomes while males possess XY pairing.
- Whether a person has XX or XY chromosomes is determined when a sperm fertilises an egg.
- Unlike the body's other cells, the cells in the egg and sperm - called gametes or sex cells - possess only one chromosome. Gametes are produced by meiosis cell division, which results in the divided cells (daughter cells) having half the number of chromosomes as the parent cells.
- All of the gametes in the mother's eggs possess X chromosomes. X chromosomes are smaller than the Y chromosomes.
- The father's sperm contains about half X and half Y chromosomes.
- The sperm are the variable factor in determining the sex of the baby.
- If the sperm carries an X chromosome, it will combine with the egg's X chromosome to form a female zygote.
- If the sperm carries a Y chromosome, it will result in a male.
- During fertilisation, gametes from the sperm combine with gametes from the egg to form a zygote.
- The zygote contains 23 pairs of chromosomes.

The above sex-inheritance in humans is summarized in the punnett square below:

PUNNETT SQUARE FOR INHERITANCE OF SEX IN HUMANS		Female chromosomes in a gamete	
		X	Y
Male chromosomes in a gamete	X	XX	XY
	Y	XY	YY

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Natural selection:

- Natural selection is defined as a process by which species of animals and plants that are best adapted to their environment survive and reproduce, while those that are less well adapted die out.
- Natural selection ensures only the fittest survive to pass their genes on to the next generation.

Example: 1

Hog deer are herbivores which live in regions of Pakistan and India. They feed on grass. Hog deer are killed and eaten by tigers.



- Hog deer are normally brown, but occasionally an albino (pure white) hog deer is born.
- The albino deers are more likely to be killed by their predator - tigers, as compared to the brown deers.

are more likely to survive. This is because they can be spotted easily against the green colour of grass. The fact that they have a colour that is not favourable for their survival shows that the albino deers are not well adapted for survival in their environment.

- Thus the albino deers are less likely to live long enough to reproduce. Hence they are less likely to pass on their genes.
- This means that natural selection is likely to ensure that very few albinos are present in a population of hog deer.
- The brown coloured hog deers are thus the fittest as they are well adapted to their environment. They survive long enough to reproduce and pass on their genes to their offspring.

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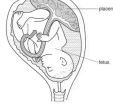
CAMBRIDGE LOWER SECONDARY
STAGE 9- BIOLOGY

CTopic: are in fetal development:

Why OR Discuss how fetal development is affected by the health of the mother, including the effect of diet, smoking and drugs.

Fetus

- Fetus is a developing baby



Pregnant

- The word pregnant refers to a woman or a female animal, having a baby developing in her body.

Pregnancy

- Pregnancy is the term used to describe the period in which a fetus develops inside a woman's womb or uterus.
- It is usually a 9 month period of pregnancy.

Placenta

- The placenta is an organ that develops in the uterus during pregnancy.
- This structure provides oxygen and nutrients to a growing baby. It also removes waste products from the baby's blood.
- The placenta attaches to the wall of the uterus, and the baby's umbilical cord arises from it.

Since substances pass to the fetus via the placenta, it becomes important to consider the effect of diet, smoking and drugs on the developing fetus.

Diet:

- A balanced diet is essential during pregnancy for the healthy development of the fetus.
- Iron is required for oxygen transport in the body.
- Increasing vitamin C intake can help increase iron absorption from foods.
- Iodine is important for normal growth and development of the baby.
- Pregnant women do need to meet calcium requirements during pregnancy.
- Pregnant women are advised to avoid foods that are associated with increased risk of the listeria bacteria (such as soft cheese and cold seafood) and to be careful with foods that are more likely to contain mercury (such as certain fish, such as fluke). Listeria can seriously affect your growing baby.
- Food items containing caffeine such as coffee, chocolate etc should be consumed in very small quantities, only as caffeine is known to slow down the growth and development of the fetus.

If poverty prevents the availability of a balanced diet to a pregnant mother, an under-weight baby might be born, that may be less capable of resisting the attack of diseases.

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Smoking

- Smoking lets nicotine and other poisons get into the mother's blood-stream. During pregnancy, smoking lets these substances pass on to the fetus. The substances present from the cigarette smoke reduce the oxygen carrying capacity and hence less oxygen reaches the fetus. As a result the fetus faces challenges in terms of survival and healthy growth. In-order to obtain more oxygen, the heart rate of the fetus increases.
- Smoking may also result in the premature birth of the baby and having a smaller weight.
- The lungs and the nervous system may also get damaged.
- Sometimes it results in a 'stillbirth', which means that the fetus dies before it is born.

Drugs:

- A drug is any substance taken into the body that modifies or effects chemical reactions in the body.
- Medicinal drugs include antibiotics that are taken to cure bacterial infections.
- Non-medicinal drugs are the misused drugs.

Alcohol

- Alcohol is classified as a drug because it affects the nervous system and slows down many actions of the body.

Effects of alcohol during pregnancy:

- Consuming even small quantities causes nerve damage.
- Consuming large quantities results in the baby being born with a smaller head, and abnormal development of eyes, nose and lips.
- Post birth, baby may suffer from poor coordination and memory, inability to concentrate well, and hyperactivity.

Non-medicinal drugs:

Examples of other non-medicinal drugs:

- Cannabis, cocaine, heroin and ecstasy

Effects of non-medicinal drugs during pregnancy:

- Slowing down of fetal development
- Fetus is less able to fight disease
- Baby may be born dependent on the drug that mother was taking during pregnancy.

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CAMBRIDGE LOWER SECONDARY
STAGE 9- BIOLOGY

Topic Environmental change and extinction:

9Bc.01 Describe what could happen to the population of a species, including extinction, when there is an environmental change.

- Organisms gradually develop features that help them survive in the environment that they live in.
- The features that organisms possess are a result of natural selection.

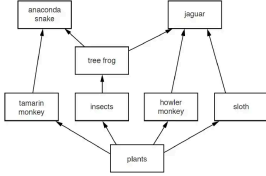
The relationship between organisms and their environment can be analysed better by studying the various food chains and food webs.

Food chain:

- A food chain describes how energy and nutrients move through an ecosystem.

Food web:

A food web is a complex network of interconnecting and overlapping food chains showing feeding relationships within a community.



Food chains and food webs help us identify the effect of an increase or decrease in the number of organisms in a habitat, meaning the changes in the population of organisms in the habitat.

Ecologists collect data about populations in two ways:

- By themselves conducting surveys of a small size of a particular habitat and then making estimates of a larger area of the same habitat based on the data collected by them.
- By using data already collected from surveys done by other researchers/ecologists.

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Population changes are brought about by a large number of factors. Some of them being:

- Predation
- Disease
- Environmental change (which might include earthquakes/ volcanoes/human activities such as fishing, farming, clearing land for other activities, mining, disposal of waste and forms of travel)

Note: Predators are organisms that hunt and kill other organisms for food

Predicting population changes:

Population changes can be predicted using reproductive rates. Reproductive rates refer to the birth rate and death rate.

Birth rate: Birth rate refers to the number of babies born per 1000 people in a country

Death rate: Death rate refers to the number of people dying per 1000 people in a country

An increase and decrease in the population:

Population is said to increase:

- If the birth rate is more than the death rate

Population is said to decrease:

- If the death rate is more than the birth rate

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Endangered species:

- An endangered species is a type of organism that is threatened by extinction.

IUCN's Red List of Threatened Species

IUCN stands for International Union for Conservation of Nature

Established in 1984, the International Union for Conservation of Nature's Red List of Threatened Species has evolved to become the world's most comprehensive information source on the global extinction risk status of animal, fungus and plant species.

The following table gives us the codes assigned for various categories into which the animals from different species are placed

Status	Symbol	Description
Endangered	EN	High risk of extinction in a part of the world or the whole world.
Critically endangered	CR	High risk of extinction in the wild
Extinct in the wild	EW	The result of various surveys show that it is no longer is present in the wild. It may survive in captivity or in collection.
Extinct	EX	No individuals have been recorded after many surveys.

As mentioned earlier, there are many causes for the endangering of species, such as loss of habitat and loss of genetic variations, pollution, etc., the main cause explained below is with regards to environmental change.

- Deforestation: Habitats are cleared for selling of trees, for growing crops, such as rubber, palm oil etc.
- Clearing land used by plants and animals for offices, factories, houses etc.
- Decrease in the amount of carbon dioxide and other due to increase in the use of vehicles
- Climate changes resulting from the burning of fossil fuels.
- Climate changes leading to an increase in the temperature of seas, increase in the sea-level, floods and droughts.
- Decrease in the number of bush and forest fires
- Pollution caused by burning of plastics, that releases toxic gases.
- Pollution by accumulation of plastics in dumping grounds and break-down of some plastics might take hundreds of years. But most plastics never ever decompose.
- Air pollution caused by gases released by factories
- Hunting has further reduced the number of already endangered species. In most cases the birth rate is less than the death rate and the species are then on the verge of extinction.

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Protecting endangered species:

Endangered species need to be prevented from extinction by increasing their numbers and by preventing a drop in their existing numbers.

This can be done in the following ways:

- Their birth rate can be raised and their death rate decreased.
- The zoo-keepers may ensure that all adult animals are healthy enough to breed. Extra care should also be provided in the rearing of the young
- Preventing predation is another way of saving the population from decreasing
- Some countries provide natural reserves for animals to live freely in the wild, at the same time banning the hunting of these animals.
- Thus increasing the birth rate, increases the number of healthy individuals who can then breed to increase the population further.

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